

Change Request: Azure Infrastructure for MCP Remote Servers

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Field	Value
Requestor	Matt Snider
Date	2026-04-03
Updated	2026-04-04
Priority	Medium
Target Completion	TBD
Environment	Production
Subscription	TBD (recommend shared services / platform subscription)
Region	East US 2 (or nearest to primary user base)

Purpose

Provision Azure infrastructure to host two MCP (Model Context Protocol) servers as remote API services, enabling organization-wide access through AI assistant integrations (Claude custom connectors). These are the first of several MCP servers that will follow the same shared infrastructure pattern.

Server 1: PlannerMCP

- **Function:** Organization-wide access to Microsoft Planner and Planner Premium (Dataverse) — 65 tools for task, plan, bucket, and project management
- **Downstream APIs:** Microsoft Graph API, Dataverse (Dynamics 365)
- **Auth model:** Entra ID delegated OAuth (On-Behalf-Of flow) — actions execute as the authenticated user
- **Container profile:** Lightweight Node.js HTTP server (~50MB image, <100MB memory at idle)

Server 2: FreshService MCP

- **Function:** Organization-wide access to FreshService ITSM — 280+ tools for tickets, assets, changes, problems, releases, knowledge base, service catalog, and operations management
- **Downstream APIs:** FreshService REST API v2

- **Auth model:** Entra ID on the Container App gates access to org users only; the FreshService API key is a shared org-wide service account credential stored in Key Vault
- **Container profile:** Lightweight Node.js HTTP server (~40MB image, <100MB memory at idle). Already has HTTP transport implemented — no code changes needed for hosting

Resources Required

1. Resource Group

Property	Value
Name	rg-mcp-servers-prod
Region	East US 2
Tags	project: mcp-servers , owner: matt.snider , environment: production

Note: This resource group will be shared by future MCP server deployments. Name it generically.

2. Azure Container Registry (ACR)

Property	Value
Name	acrmcpservers (or per org naming convention)
Resource Group	rg-mcp-servers-prod
SKU	Basic
Admin User	Disabled
Public Network Access	Enabled (required for GitHub Actions CI/CD)

Purpose: Stores Docker container images for all MCP servers. Basic SKU is sufficient (10GB storage, 2 webhooks).

3. Log Analytics Workspace

Property	Value
Name	log-mcp-servers
Resource Group	rg-mcp-servers-prod

Property	Value
Retention	30 days (default)

Purpose: Required by Container Apps Environment. Receives container logs, system logs, and request metrics.

4. Azure Container Apps Environment

Property	Value
Name	cae-mcp-servers
Resource Group	rg-mcp-servers-prod
Log Analytics Workspace	log-mcp-servers
Zone Redundancy	Disabled (cost optimization; can enable later)
Internal Only	No — must be externally accessible

Purpose: Shared hosting environment for all MCP server Container Apps. This is a managed Kubernetes environment — no cluster management required.

5. User-Assigned Managed Identity

Property	Value
Name	id-mcp-servers
Resource Group	rg-mcp-servers-prod

Role Assignments Required:

Role	Scope	Purpose
AcrPull	acrmcpservers (Container Registry)	Pull container images at deployment
Key Vault Secrets User	kv-mcp-servers (Key Vault)	Read secrets at runtime

6. Azure Key Vault

Property	Value
Name	kv-mcp-servers (or per org naming convention)
Resource Group	rg-mcp-servers-prod

Property	Value
SKU	Standard
Soft Delete	Enabled (default)
Purge Protection	Enabled
RBAC Authorization	Enabled (use Azure RBAC, not vault access policies)

Initial Secrets to Create:

Secret Name	Value	Source
<code>planner-mcp-client-secret</code>	<i>(provided after Entra app registration — see Section 9A)</i>	Entra ID App Registration
<code>freshservice-mcp-client-secret</code>	<i>(provided after Entra app registration — see Section 9B)</i>	Entra ID App Registration
<code>freshservice-api-key</code>	<i>(provided by Matt Snider — existing FreshService API key)</i>	FreshService Admin Console
<code>freshservice-domain</code>	<i>(provided by Matt Snider — e.g., <code>yourcompany</code>)</i>	FreshService tenant subdomain

Access: - `id-mcp-servers` managed identity: `Key Vault Secrets User` role - Requestor (Matt Snider): `Key Vault Administrator` role (for secret management)

7. Azure Container App — PlannerMCP

Property	Value
Name	<code>ca-planner-mcp</code>
Resource Group	<code>rg-mcp-servers-prod</code>
Environment	<code>cae-mcp-servers</code>
User-Assigned Identity	<code>id-mcp-servers</code>

Container Configuration:

Property	Value
Image	<code>acrmcpservers.azurecr.io/planner-mcp:latest</code>
CPU	0.5 vCPU
Memory	1 Gi

Property	Value
Min Replicas	1
Max Replicas	5
Scale Rule	HTTP concurrent requests (10 per replica)

Note: Image will not exist until the development team pushes the first build. The Container App can be created with a placeholder image (`mcr.microsoft.com/k8se/quickstart:latest`) and updated later.

Ingress Configuration:

Property	Value
Ingress	External
Target Port	8080
Transport	HTTP (TLS terminated at ingress)
Allow Insecure	No

Environment Variables:

Name	Value	Source
TENANT_ID	508bd243-a211-474f-9fc9-83dea3cc95c7	Plaintext
CLIENT_ID	(from Entra app registration — see Section 8)	Plaintext
CLIENT_SECRET	Key Vault reference	kv-mcp-servers / planner-mcp-client-secret
PORT	8080	Plaintext
NODE_ENV	production	Plaintext

Secrets (Container App level):

Secret Name	Source
planner-mcp-client-secret	Key Vault reference: <code>https://kv-mcp-servers.vault.azure.net/secrets/planner-mcp-client-secret</code> , identity: <code>id-mcp-servers</code>

Health Probes:

Probe	Type	Path	Port	Period	Failure Threshold
Liveness	HTTP GET	/health	8080	30s	3
Readiness	HTTP GET	/health	8080	10s	3
Startup	HTTP GET	/health	8080	5s	10

8. Azure Container App — FreshService MCP

Property	Value
Name	ca-freshservice-mcp
Resource Group	rg-mcp-servers-prod
Environment	cae-mcp-servers
User-Assigned Identity	id-mcp-servers

Container Configuration:

Property	Value
Image	acrmcpservers.azurecr.io/freshservice-mcp:latest
CPU	0.5 vCPU
Memory	1 Gi
Min Replicas	1
Max Replicas	5
Scale Rule	HTTP concurrent requests (10 per replica)

Note: Image will not exist until the development team pushes the first build. The Container App can be created with a placeholder image (mcr.microsoft.com/k8se/quickstart:latest) and updated later.

Ingress Configuration:

Property	Value
Ingress	External
Target Port	8080
Transport	HTTP (TLS terminated at ingress)

Property	Value
Allow Insecure	No

Environment Variables:

Name	Value	Source
FRESHSERVICE_API_KEY	Key Vault reference	kv-mcp-servers / freshservice-api-key
FRESHSERVICE_DOMAIN	Key Vault reference	kv-mcp-servers / freshservice-domain
MCP_TRANSPORT	http	Plaintext
PORT	8080	Plaintext
NODE_ENV	production	Plaintext

Secrets (Container App level):

Secret Name	Source
freshservice-api-key	Key Vault reference: https://kv-mcp-servers.vault.azure.net/secrets/freshservice-api-key , identity: id-mcp-servers
freshservice-domain	Key Vault reference: https://kv-mcp-servers.vault.azure.net/secrets/freshservice-domain , identity: id-mcp-servers

Health Probes:

Probe	Type	Path	Port	Period	Failure Threshold
Liveness	HTTP GET	/health	8080	30s	3
Readiness	HTTP GET	/health	8080	10s	3
Startup	HTTP GET	/health	8080	5s	10

Note: The FreshService MCP does not require its own Entra ID App Registration — it does not use delegated OAuth. Instead, access is gated by enabling Entra ID built-in authentication on the Container App itself (see Section 10). The FreshService API key is a shared org-wide service account credential; all users share the same downstream API access. Per-user identity is enforced at the Container App ingress level via Entra ID.

Authentication — Entra ID Built-In Auth on Container App:

Enable Container Apps built-in authentication (EasyAuth) for `ca-freshservice-mcp`:

Property	Value
Authentication	Enabled
Identity Provider	Microsoft Entra ID
Action for unauthenticated requests	Return 401
App Registration	Use the shared <code>MCP - FreshService</code> app registration (see Section 9B)

This ensures only authenticated users in your Entra ID tenant can reach the FreshService MCP endpoint. The MCP server itself handles downstream FreshService API calls using the shared API key from Key Vault.

9A. Entra ID App Registration — PlannerMCP

Property	Value
Name	<code>MCP - Microsoft Planner</code>
Supported Account Types	Single tenant (this organization only)
Application ID URI	<code>api://<generated-client-id></code>

Redirect URIs (Web):

URI	Purpose
<code>https://claude.ai/api/mcp/auth_callback</code>	Claude connector OAuth callback
<code>https://ca-planner-mcp.<region>.azurecontainerapps.io/auth/callback</code>	Direct server callback (testing)

Note: The Container App FQDN redirect URI should be updated after the Container App is provisioned and the actual FQDN is known.

API Permissions (Delegated — NOT Application):

API	Permission	Type	Admin Consent Required
Microsoft Graph	<code>Tasks.ReadWrite</code>	Delegated	No
Microsoft Graph	<code>Group.ReadWrite.All</code>	Delegated	Yes

API	Permission	Type	Admin Consent Required
Microsoft Graph	GroupMember.Read.All	Delegated	Yes
Microsoft Graph	Team.ReadBasic.All	Delegated	No
Microsoft Graph	User.Read	Delegated	No
Microsoft Graph	User.ReadBasic.All	Delegated	No
Dynamics CRM	user_impersonation	Delegated	No

Action Required: Grant admin consent for all permissions after adding them.

Expose an API:

Property	Value
Application ID URI	api://<client-id>
Scope Name	MCP.Access
Scope Display Name	Access MCP - Microsoft Planner
Who can consent	Admins and users
State	Enabled

Certificates & Secrets:

Type	Description	Expiry
Client Secret	PlannerMCP server credential	12 months (set calendar reminder for rotation)

Action Required: After creating the client secret, store the value in Key Vault as `planner-mcp-client-secret` (see Section 6).

Token Configuration (Optional Claims):

Token Type	Claim	Purpose
ID Token	email	Audit logging
ID Token	preferred_username	Audit logging

9B. Entra ID App Registration — FreshService MCP

This is a simpler registration — it only gates access to the Container App. No downstream API permissions are needed because FreshService uses its own API key.

Property	Value
Name	MCP – FreshService
Supported Account Types	Single tenant (this organization only)
Application ID URI	api://<generated-client-id>

Redirect URIs (Web):

URI	Purpose
https://claude.ai/api/mcp/auth_callback	Claude connector OAuth callback
https://ca-freshservice-mcp.<region>.azurecontainerapps.io/auth/callback	Direct server callback (testing)

Note: The Container App FQDN redirect URI should be updated after the Container App is provisioned and the actual FQDN is known.

API Permissions (Delegated):

API	Permission	Type	Admin Consent Required
Microsoft Graph	User.Read	Delegated	No

Only User.Read is needed — this registration is used solely to authenticate the user’s identity at the Container App level. No downstream Microsoft API calls are made.

Expose an API:

Property	Value
Application ID URI	api://<client-id>
Scope Name	MCP.Access
Scope Display Name	Access MCP - FreshService
Who can consent	Admins and users
State	Enabled

Certificates & Secrets:

Type	Description	Expiry
Client Secret	FreshService MCP Container App auth credential	12 months (set calendar reminder for rotation)

Action Required: After creating the client secret, store the value in Key Vault as `freshservice-mcp-client-secret` (see Section 6). This secret is used by the Container App’s built-in authentication — not by the FreshService MCP application code.

Token Configuration (Optional Claims):

Token Type	Claim	Purpose
ID Token	<code>email</code>	Audit logging
ID Token	<code>preferred_username</code>	Audit logging

10. Diagnostic Settings

Configure diagnostics on **both** Container Apps (`ca-planner-mcp` and `ca-freshservice-mcp`) to stream to Log Analytics:

Log Category	Enabled
<code>ContainerAppConsoleLogs</code>	Yes
<code>ContainerAppSystemLogs</code>	Yes

Metric Category	Enabled
<code>AllMetrics</code>	Yes

Destination: `log-mcp-servers` Log Analytics Workspace

11. IP Restrictions (Optional — Recommended for Production)

Restrict inbound traffic on **both** Container Apps’ ingress to:

Rule Name	IP Range / CIDR	Action	Priority
<code>anthropic-cloud</code>	<i>(Request from Anthropic support)</i>	Allow	100
<code>corp-egress</code>	<i>(Corporate VPN/proxy egress IPs)</i>	Allow	200
<code>deny-all</code>	<code>0.0.0.0/0</code>	Deny	1000

Note: IP restrictions can be added after initial deployment and testing. The MCP server must be reachable from Anthropic’s cloud infrastructure — all Claude traffic originates from their servers, not from end-user devices. Contact Anthropic support for their current IP ranges, or leave open initially and restrict later.

Provisioning Summary

#	Resource	Type	Name
1	Resource Group	Microsoft.Resources/resourceGroups	rg-mcp-servers-prod
2	Container Registry	Microsoft.ContainerRegistry/registries	acrmcpservers
3	Log Analytics Workspace	Microsoft.OperationalInsights/workspaces	log-mcp-servers
4	Container Apps Environment	Microsoft.App/managedEnvironments	cae-mcp-servers
5	Managed Identity	Microsoft.ManagedIdentity/userAssignedIdentities	id-mcp-servers
6	Key Vault	Microsoft.KeyVault/vaults	kv-mcp-servers
7	Container App	Microsoft.App/containerApps	ca-planner-mcp
8	Container App	Microsoft.App/containerApps	ca-freshservice-mcp
9	App Registration	Entra ID	MCP – Microsoft Planner
10	App Registration	Entra ID	MCP – FreshService

Estimated Monthly Cost

Resource	Estimated Cost
Container App — PlannerMCP (0.5 vCPU, 1Gi, 1 min replica)	\$30–50

Resource	Estimated Cost
Container App — FreshService MCP (0.5 vCPU, 1Gi, 1 min replica)	\$30–50
Container Registry (Basic, shared)	\$5
Key Vault (Standard, low volume)	<\$1
Log Analytics (30-day retention, low volume)	\$5–10
Total	~\$70–115/month

Cost will increase proportionally as additional MCP servers are added to the same environment. The Container Apps Environment, ACR, Key Vault, and Log Analytics are shared — incremental cost per additional MCP server is ~\$30-50/month (Container App only).

What the Development Team Needs Back

After provisioning, please provide the following for **each** server:

PlannerMCP:

Item	Example
Container App FQDN	ca-planner-mcp.nicemeadow-abc12345.eastus2.azurecontainerapps.io
Entra App Client ID	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx
Entra App Client Secret	(store in Key Vault as <code>planner-mcp-client-secret</code> , confirm done)

FreshService MCP:

Item	Example
Container App FQDN	ca-freshservice-mcp.nicemeadow-abc12345.eastus2.azurecontainerapps.io
Entra App Client ID	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx
Entra App Client Secret	(store in Key Vault as <code>freshservice-mcp-client-secret</code> , confirm done)
FreshService secrets confirmed	(confirm <code>freshservice-api-key</code> and <code>freshservice-domain</code> stored in Key Vault)

Shared Resources:

Item	Example
ACR Login Server	acrmcpservers.azurecr.io
Key Vault URI	https://kv-mcp-servers.vault.azure.net
Managed Identity Client ID	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
Managed Identity Resource ID	/subscriptions/.../id-mcp-servers

Future Additions (No Action Now)

The following MCP servers will be deployed to the same environment in the coming weeks. Each will be a new Container App + Entra App Registration, reusing the shared ACR, Key Vault, Log Analytics, and Container Apps Environment:

- Azure Cost Management (ca-azure-billing-mcp)
- ControlUp DEX (ca-controlup-mcp)
- Microsoft 365 (ca-m365-mcp)
- Nerdio AVD Manager (ca-nerdio-mcp)
- Evisort Contracts (ca-evisort-mcp)

No infrastructure changes needed for these — just additional Container Apps and App Registrations following the same pattern.

Questions / Dependencies

1. **Subscription selection** — Which subscription should host these resources? Recommend a shared services or platform subscription rather than a workload-specific one, since this will host 7 services.
2. **Naming convention** — Resource names above follow {type}-mcp-{service} pattern. Adjust to match organizational naming standards if different.
3. **Region** — East US 2 is proposed. Confirm this aligns with your data residency and latency requirements.
4. **Entra admin consent** — Group.ReadWrite.All and GroupMember.Read.All require admin consent. Confirm who will grant this.
5. **ACR public access** — Required for GitHub Actions CI/CD. If org policy requires private ACR, we'll need a self-hosted runner or ACR Tasks (noted in Section 2).